

amazon



Web Scraper

Prayuja Teli.
pst29@drexel.edu

Introduction and Purpose of Dataset

This project aims to create a web scraper to gather product data from Amazon. The dataset includes product descriptions, prices, ratings, review counts, and URLs. The primary purpose is to enable data analysis on e-commerce products to identify trends, compare pricing, and evaluate product popularity.

Potential Users and Applications

- Consumers
- Data analysts and Market Researchers
- Retailers and Small Businesses
- Developers

Source of Data

The data is scraped from Amazon's search result pages using Selenium and BeautifulSoup. The source URL is dynamically generated based on user-defined search terms.

Approach to Acquiring the Data

Web Scraping Framework:

- **Selenium:** Automates browsing and interacts with Amazon's dynamic web pages.
- **BeautifulSoup:** Parses the HTML content to extract product details.

Approach to Preprocessing Data

- Remove rows containing missing or invalid values (e.g., 'N/A').
- Convert fields like price and review count to numeric formats.
- Generate a summary table for data exploration.
- Cleaned data is saved in a `cleaned\_results.csv` file for further analysis.

Distribution Approach

- The scraped and cleaned data is distributed in CSV format.
- **Single File Output:** All records consolidated into `all\_results.csv` or `cleaned\_results.csv`.
- **Visualization Outputs:** Include histograms, box plots, and correlation heatmaps saved as image files for analysis insights.
- **Optional:** Results can be saved as individual CSV files for specific search terms.

Discussion of Access Rights

Publicly available dataset so restricting code or logic won't be much of use.

Issues and Limitations

- **Dynamic Content Challenges:** Amazon's use of JavaScript and dynamic loading requires Selenium for effective scraping.
- **Anti Scraping Measures:** Potential rate-limiting or IP bans if the scraping frequency is too high.
- **Incomplete Data:** Some fields (e.g., price, rating) might not be available for all products, resulting in missing values.
- **Scalability:** Limited to Amazon's structure and might require modifications for other e-commerce platforms.

Team and Contributions

This project is an individual effort, where I developed and implemented:

- **Web Scraping Logic:** Using Selenium and BeautifulSoup for efficient data extraction.
- **Data Cleaning Pipeline:** To ensure high-quality datasets for analysis.
- **Data Visualization:** Using libraries like Matplotlib and Seaborn to generate insightful plots.

Thank You!